

Take-Home Assignment: Custom Masked Attention in JAX Pallas for RecSys

Context

You are a RecSys ML platform engineer optimizing attention for sequential recommendation models on GPU.

Task

Implement a custom masked multi-head attention forward pass using JAX Pallas GPU backend.

Reference: <https://github.com/jax-ml/jax/blob/main/jax/experimental/pallas/ops/gpu/attention.py>

Inputs

Q, K, V each shaped [batch, seq_len, num_heads, head_dim], where seq_len = history_len + candidate_len

Custom Mask

- First history_len positions (history) attend bidirectionally to each other
- Last candidate_len positions (candidates) attend only to the history part (no attention among candidates)

Requirements

- Implement the mask correctly in the Pallas kernel (pass history_len and candidate_len as parameters)
- Assume history_len and candidate_len are the same across the batch (fixed padding)
- Use numerically stable softmax (FlashAttention-style)
- **Performance optimization:** Leverage the mask structure (history bidirectional + candidates only attend to history) to reduce unnecessary computation and memory access. Consider ways to avoid wasteful cycles in fully masked regions through tiling strategy, loop bounds, or data loading decisions.

Test

Validate correctness with sample data (e.g., batch=4, seq_len=1024, history_len=512, heads=8, head_dim=64)

Note on Hardware

You do not need a personal GPU. CPU execution is acceptable for correctness verification (just set `jax.config.update("jax_platform_name", "cpu")`). If you have access to a GPU (e.g., via Colab free tier), feel free to use it for performance experiments.

Deliverables

- Jupyter notebook: full code, correctness tests, performance discussion
- Short write-up (≤1 page): design choices, mask-aware optimization ideas, expected benefit

AI Usage Note

You may (and are encouraged to) use AI tools. Core ideas must reflect your understanding. Briefly mention in write-up: (1) which parts used AI, (2) key prompts/changes, (3) your reasoning.

Time Estimate

4–6 hours

Submission

GitHub repo link (private)

Bonus (Optional)

- Sketch support for per-sample variable history_len
- Rough benchmark (naive vs optimized)